

LARGE-SCALE STRUCTURE

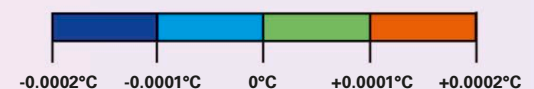
A computer simulation of the large-scale structure of the universe shows the concentration of matter into filaments across a billion light-years of space. According to models of cosmic evolution, dark matter (yellow) formed clumps around which visible matter (lighter pinks) coalesced. Clusters of galaxies are concentrated at nodes where filaments meet; superclusters are strung out along the filaments.

THE ORIGIN OF SUPERCLUSTERS

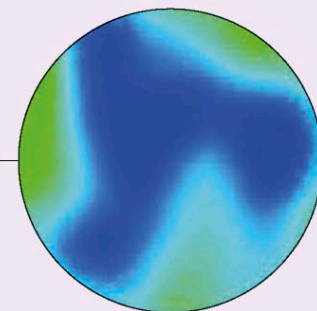
The sheer size of galaxy superclusters makes their origins important in understanding the structure and properties of the universe as a whole. Theoretically, superclusters could either have formed from the gradual coalescence of matter pulled together by gravity alone, or they could mark large-scale concentrations of matter that were present from the earliest times and within which galaxies and galaxy clusters subsequently developed. The first detailed maps of the cosmic microwave background radiation (CMBR), produced by the COBE satellite in 1992, provided evidence that large-scale structures were present from the earliest times, confirming the second theory. The structures themselves are thought to have originated as microscopic variations in temperature and density in the early universe and then enlarged to enormous scale by cosmic expansion, eventually giving rise to the “Swiss cheese” distribution of matter in the universe today.

RADIATION FROM THE SKY

This elliptical map shows tiny variations in the temperature of the cosmic microwave background radiation (CMBR) over the whole sky, as measured by the Wilkinson Microwave Anisotropy Probe (WMAP). The variations—anisotropies—are color-coded according to the temperature scale shown below.

TEMPERATURE VARIATION IN THE CMBR

the red band running across the center of the map is caused by microwaves from the Milky Way

**THE GREAT COLD SPOT**

This large cold spot in the CMBR may be due to a huge void around 6–10 billion light-years away.

SEEING THE FIRST LIGHT

The cosmic background microwave radiation detected by WMAP consists of photons that “escaped” from matter less than 380,000 years after the Big Bang. This was after the first atoms formed but before the first stars ignited—a period known as the Dark Ages. In contrast, Hubble can observe back only to about 400–800 million years after the Big Bang, when early galaxies had formed.

